STRATEGIES TO IMPROVE THE RECOGNITION AND WORKUP OF ABUSIVE HEAD TRAUMA (AHT) IN INFANTS AND CHILDREN

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LEARNING OBJECTIVES

- Describe the types of injuries which cause suspicion for abusive head trauma
- Explain the appropriate workup to initiate if a physician suspects abusive head trauma
- Describe the threshold for reporting cases to the Department of Children and Family Services (DCFS) and take the appropriate action
- Describe the disparities and implicit biases which impact screening, workup and reporting abusive head trauma

CHILD ABUSE EVALUATION

- Purpose:
  - To assess for alternative diagnoses to explain symptoms
  - Detect occult injuries
- Goal is not to rule out abuse
- Remember to assess for other children in the home
- Most families need ongoing injury surveillance and additional resources/support

ABUSIVE HEAD TRAUMA OVERVIEW

- Epidemiology and Background
- Clinical Presentation
- Evaluation
- Differential Diagnosis
- Pathophysiology
- Outcomes
- Reporting to DCFS and Court overview
- Resources

EPIDEMIOLOGY

- 32-38 cases/100,000 children in first year of life
  - 95% of all serious intracranial injuries and 64% of all head injuries in infants <1 year are due to physical abuse
- Leading cause of death from trauma in children < 2 years
- Median age of victims is 2-6 months
  - Overlaps the peak ages of normal infant crying

TERMINOLOGY

- AAP and Committee of Child Abuse and Neglect in 2009: Recommended term "Abusive Head Trauma" (AHT)
- More inclusive since cause is usually multifactorial
  - Shaking
  - Shaking and impact
  - Impact

NO CONTROVERSY

- Although, the courtroom has become a place where speculative theories are often presented as legitimate diagnoses
  - No reliable medical evidence that cerebral sinovenous thrombosis, hypoxic-ischemic injury, lumbar puncture or dysphagic choking/vomiting causes constellation of injuries seen in AHT
  - Or that asymptomatic birth-related subdural hemorrhage can result in rebleeding and collapse
- Narang SK, et al. Acceptance of shaken baby syndrome and abusive head trauma as medical diagnoses. *J Pediatr*. 2016.
  - 93% of 600+ physicians surveyed believe that AHT is a valid evidence-based diagnosis
- **Choudhary AK, et al. Consensus statement on abusive head trauma in infants and young children. Pediatric Radiology. 2018**
  - Supported by Society for Pediatric Radiology, American Society of Pediatric Neuroradiology, American Academy of Pediatrics, etc.

KEY HISTORICAL POINTS IN OBTAINING TRAUMA HISTORY

- Trauma history is often missing or misleading
- Detailed description of:
  - Mechanism of injury: surface, height, position found
  - Appearance of signs and symptoms, as well as response by caregiver
  - Witnesses present during injury
- Timing of injury → When was child last normal?
- Obtain history from child and multiple caregivers
- Birth history
- Developmental history→Delays place at increased risk
- Past medical history→Previous trauma, chronic illness
- Social and family history

HISTORY RED FLAGS

- No history of trauma
- Inconsistent, changing, or conflicting
- History of trauma inconsistent with mechanism provided
  - Short fall (<4 feet)
- Inconsistent with child's development

UNEXPLAINED HEAD TRAUMA

- Jenny, et al. 1999 - Nearly 1/3 of infants with inflicted head injury were missed on initial presentation for medical care
  - 25% were then reinjured before they were diagnosed with physical abuse
  - 10% of missed cases suffered fatal injuries
- Adamsbaum, et al. 2010 - In study where perpetrators admitted to shaking a child, more than ¼ of infants had presented with poor wt gain, bruising, and fractures
  - Abuse was not suspected
- Chadwick, et al. 2008 - Annual risk of dying from a short fall (<1.5 meters) is < 0.48 in 1 million in children under 5 years old

INITIAL PRESENTATION

- Severe signs – Seen in ~65%
  - Seizures, apnea, coma, altered mental status, cardiopulmonary arrest
- Nonspecific signs
  - Feeding difficulty, vomiting, breathing problems, lethargy, lump on head, BRUE, large or enlarging head circumference
- May have no signs or symptoms

KEY PHYSICAL EXAM POINTS

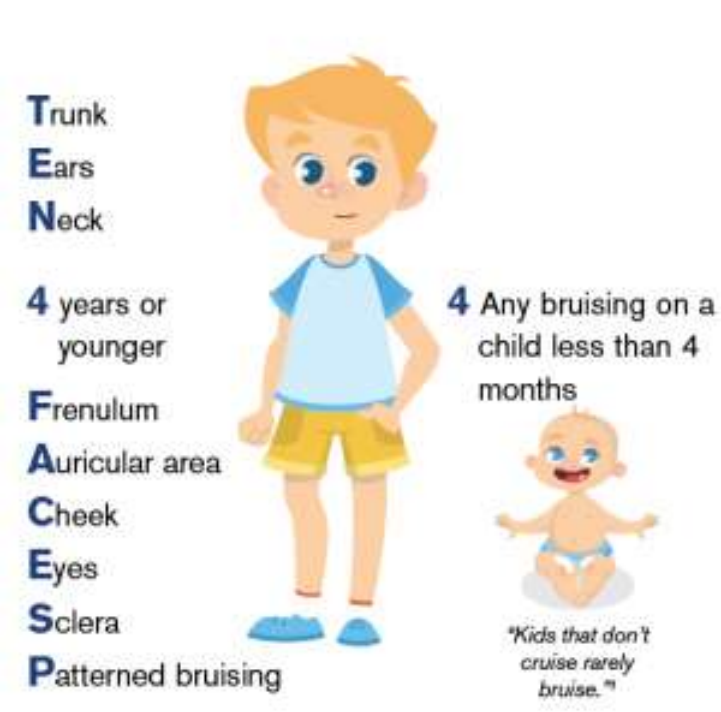
- Growth parameters, especially weight, head circumference
- Perform complete, **unclothed** exam
- Description of bruising/skin trauma
  - Size, location, shape, color, palpable/flat
- Areas not to miss
  - Scalp
  - Ears
  - Oral cavity, including teeth and frenulum
  - Back/buttocks
  - Inner thighs
  - Genitals
- Photodocumentation

BRUISING IN PRE-AMBULATORY INFANT

- If you don't cruise, you don't bruise
- Any bruising in non-mobile infant should prompt full child abuse evaluation and report to DCFS
- Bruising can be a sentinel event
  - Subtle, often overlooked sign
  - Warning that child needs further assessment

RED FLAGS IN BRUISING

- **TEN-4 criteria:**
- **Any bruising in pre-cruising child** or bruising of **torso, ears, or neck in child <4 years**
- **97% sensitive, 84% specific**



Pierce MC, et al. Bruising characteristics discriminating physical child abuse from accidental trauma. Pediatrics. 2010;125:67-74.

EAR BRUISING

TORN FRENULUM

cpdt.org.uk

SUBCONJUNCTIVAL HEMORRHAGES

DON'T FORGET ABOUT THE SIBLINGS

- At increased risk since they share a home with an abused child
- Lindberg, et al. 2012 - When skeletal surveys were obtained in siblings, <24 months old, of an index child
  - At least one abusive fracture was found in 16 of 134 contacts (11.9%)
  - No indication of these fractures on physical exam
- Twins are higher risk of abusive fracture compared to non-twin contacts
  - Stress of caring for twins, prematurity, developmental delays

ABUSIVE HEAD TRAUMA CLINICAL FINDINGS

- Possible findings can include:
  - Bruising (but **often no external findings**)
  - Rib or long bone fractures
  - Abdominal trauma
  - Eye trauma
  - Intracranial hemorrhage (speficially subdural hemorrhage) or contusions
  - Signs of external head trauma, e.g. swelling, bruising
- Maguire et al., 2011 - Study of 1000 children, <3 yrs old with intracranial injury estimated probability injury related to AHT based on 6 findings
  - Apnea, retinal hemorrhages, rib fractures, long-bone fractures, seizures, and head bruising, neck bruising, or both
  - PPV for AHT is 4% if none of the findings are present; **97% if all six findings are present**

EVALUATION

- Multidisciplinary approach
  - Child abuse team
  - Social work
  - Radiology
  - Neurology
  - Neurosurgery
  - Ophthalmology
  - Trauma

FRACTURES

- Most common fractures in AHT:
  - Rib fractures
  - Classic metaphyseal lesions (corner or bucket-handle fractures)
  - Seldom show clinical signs
- Skull fractures also can be found

FRACTURES

- No single fracture is pathognomonic for abuse
- High specificity
  - Classic metaphyseal lesions (CML's)
  - Rib fractures, especially posterior
  - Scapular, sternal, and spinous process fractures
- Moderate specificity
  - Multiple fractures, especially bilateral
  - Fractures of different ages
  - Epiphyseal separations
  - Vertebral body fractures and subluxations
  - Digital fractures
  - Complex skull fractures
    - More than one fx line (stellate or egg-shell fx), cross suture lines, comminuted (multiple pieces), depressed, widened fx margin (>3mm), occipital fx

SKELETAL SURVEY

- Indicated in any child under 2 years of age with suspicious or unexplained injuries
- Recommendations given by AAP and ACR
  - Include oblique ribs
- Acute rib fractures are difficult to visualize and require follow-up survey 2 weeks after initial survey

RIB FRACTURES

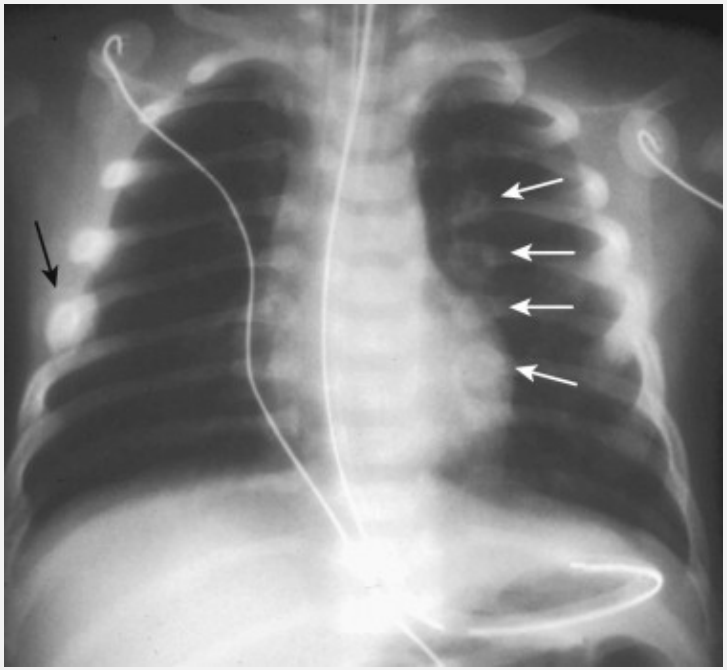
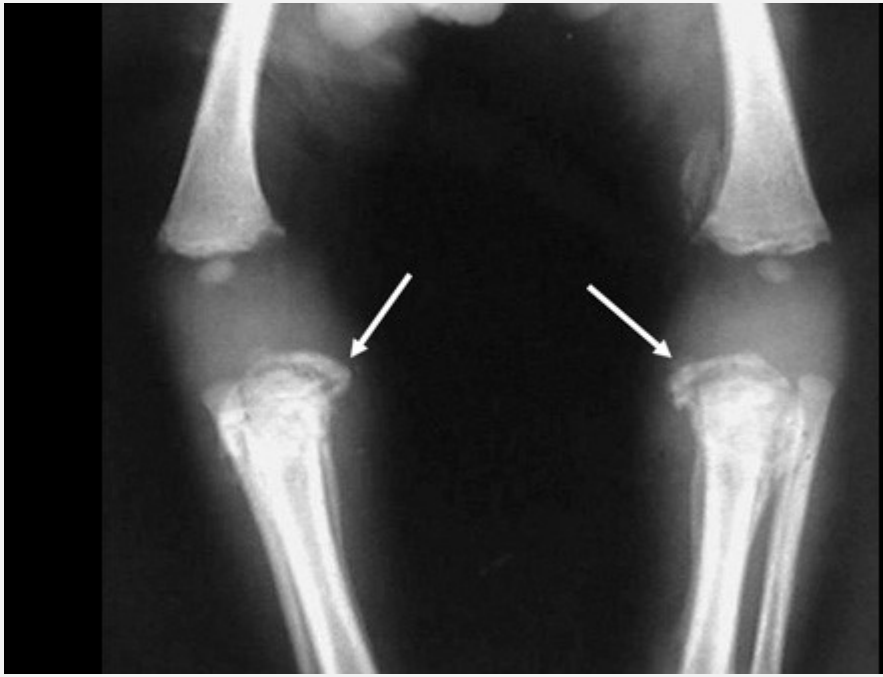


Image: Jenny, C. Child Abuse and Neglect: Diagnosis, Treatment, and Evaluation. 2011. Elsevier, St. Louis, Missouri. Ch. 33.

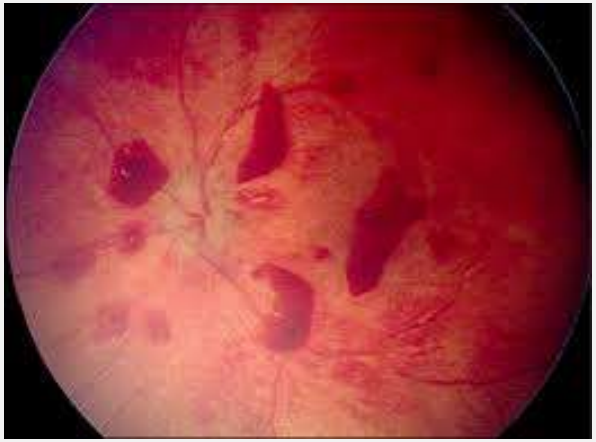
CLASSIC METAPHYSEAL LESION



http://img.medscape.com/pi/features/slideshow-slide/child-abuse/fig10.jpg

http://www.urmc.rochester.edu/radiology/education/materials/interesting_cases/images/IS_Case462/IS_Case462_Fig3.jpg

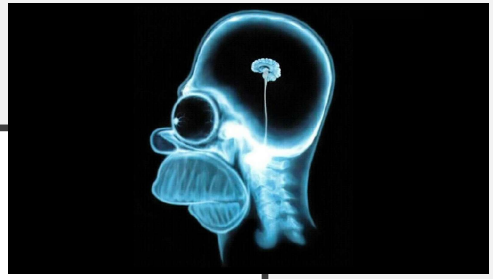
RETINAL HEMORRHAGES



- Dilated fundoscopic exam by ophthalmologist ASAP
  - Complete description includes layers involved, location, number
- **Mandatory for all infants with significant intracranial hemorrhage on head CT**
- Due to torsional forces in shaking
- Broad differential diagnosis
- Highly specific for abuse
  - **Too numerous to count, multiple retinal layers with extension to periphery**
  - **Retinoschisis, macular folds**
- Cannot be dated
- Occurs in ~85% of AHT
  - Can be unilateral or bilateral
  - In contrast, found in <3% of patients evaluated for accidental head injury
    - Focal, in posterior pole, superficial layers of the retina

Image: http://www.anatomybox.com/retinal-hemorrhages/

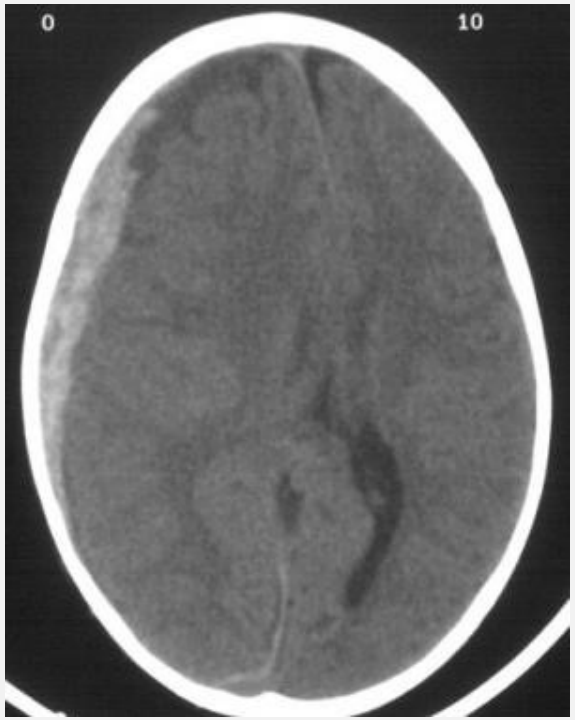
NEUROIMAGING



- Head CT
  - Strongly consider in infants with other abusive injuries or high suspicion for intracranial injury
  - 3D-reconstruction of the bone windows helpful to determine fractures vs. sutures, areas of impact
- Brain MRI
  - Follow-up study for children with abnormal head CT, needed to evaluate parenchymal injury (contusions, hypoxic-ischemic changes, diffuse axonal injury)
  - Consider if head CT normal but still concerns regarding intracranial injury, e.g. rapidly increasing head circumference
- Neck/Spinal cord imaging
  - Should be completed in infants with head injury concerning for abusive head trauma
  - Injury more likely to be soft-tissue or ligamentous, can have subdural hematoma track from posterior fossa

SUBDURAL HEMORRHAGE

- Most commonly observed intracranial lesion in young infants with AHT (in up to 90%)
- May have mixed attenuation from rupture of arachnoid membrane from mixing CSF
- Precise estimation of age of mixed attenuation subdural on initial Head CT should be avoided



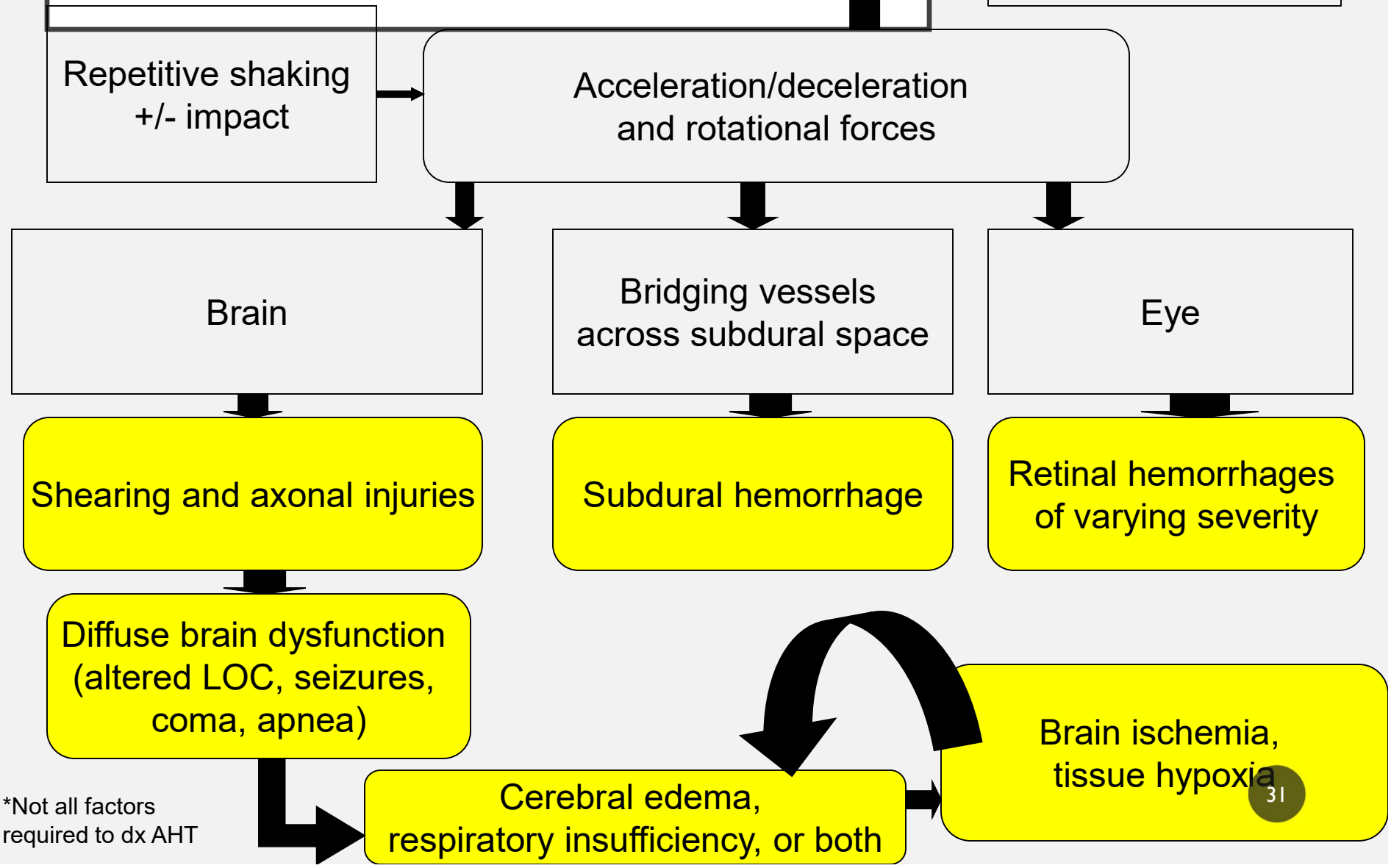
DIFFERENTIAL DIAGNOSIS INTRACRANIAL HEMORRHAGE

- Birth trauma
  - Most retinal hemorrhage/ICH resolve by 4-6 weeks of age
- Accidental trauma
  - Fall of multiple stories, fall down stairs, MVC, crush injury
  - Typical picture of short fall includes possible linear skull fracture, epidural hemorrhage, focal contusion, small focal subdural or subarachnoid hemorrhage
- Abusive head trauma
- Glutaric aciduria type I
  - Frontotemporal atrophy, developmental delay
- Vitamin K deficiency
- Benign Enlargement of Subarachnoid Space (BESS)
  - Self-limited, resolves by 2-3 years, normal development
  - May predispose to frontal SDH with minor trauma
- Bleeding disorder
- Vascular anomaly

PATHOPHYSIOLOGY

- Few direct observations
- What we do know is based on:
  - Perpetrator confessions
    - Violent shaking is most commonly reported mechanism
    - 55% reported separate, repetitive incidents
  - Computer and animal models

PATHOPHYSIOLOGY

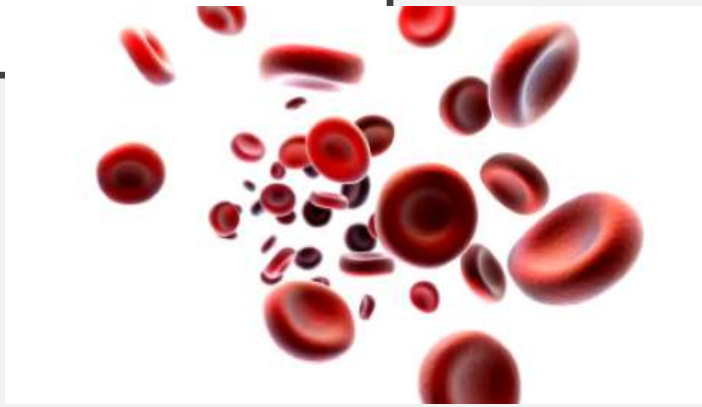


*Not all factors required to dx AHT

ABDOMINAL IMAGING

- Abdominal CT
  - No ultrasound
- Indications:
  - History of abdominal injury
  - Signs and symptoms of abdominal injury
  - Elevated AST, ALT>80
- Intentional blunt abdominal trauma has second highest mortality rate in abusive injuries

INITIAL LABORATORY EVALUATION



- Bruising
  - CBC with differential, PT/aPTT/INR, vW panel
- Intracranial bleeding (e.g. RH, SDH)
  - CBC with differential, PT/aPTT/INR, factor VIII, factor IX, DIC panel (D-dimer, fibrinogen)
- Multiple fractures
  - Ca, Phosphorus, Alkaline phosphatase, iPTH, 25-hydroxy-Vitamin D
- Abdomen
  - ALT, AST, amylase, lipase, Hgb, urinalysis

Image: www.womenshealthency.com

Anderst JD. Evaluation for bleeding disorders in suspected child abuse. Pediatrics. 2013;131:e1314-e1322.

OUTCOMES IN AHT

- Spastic hemiplegia or quadriplegia (15-64%)
- Microcephaly with cortico-subcortical atrophy (61-100%)
- Visual impairment (18-48%)
- Intractable epilepsy (11-32%)
- Language disorders (37-64%)
- Agitation, aggression, tantrums, attention deficits, memory deficits, inhibition or initiation deficits (23-59%)
- Death (20-25%)

LOS ANGELES DCFS HOTLINE

- Call if you are suspicious for abuse or neglect
- Open 24/7→**1-800-540-4000**
- You are a mandated reporter→Reporting may be life saving
- Investigation can include a home visit, obtaining additional information from neighbors/teachers/family, and/or drug testing of family members
- Law enforcement is also cross-reported
- Any caregiver can momentarily lose control and injure a child
- Caregivers other then parents may have injured child

TALKING WITH THE FAMILY

- Safety of infant is our priority and legal mandate
- **Risk of missing abuse is potentially life threatening**
- **You do not have to diagnose abuse, only have a reasonable suspicion, to report**
- Inform family in non-judgmental style that reporting is required
  - "It is our protocol when young children have this type of injury that it be reported to DCFS"
- Continue to keep family updated with results of diagnostic studies

CULTURAL BIAS

- Hymel 2018 – Children with diverse racial and ethnic backgrounds with head injuries are **almost twice as likely** to be reported and evaluated for AHT than white children with similar symptoms
- Lane, et al. 2002 – Minority children were more likely to have a skeletal survey completed and be reported to child protective services
- Systemic issues
  - Bilingual providers, Interpreters
  - Structural racism (e.g. exclusion of Black families from home ownership) → Poverty
    - Increased exposure to social services systems → Surveillance bias

ROLE OF THE PEDIATRICIAN

- Identify medical diagnoses, evaluate and treat child's injuries
- Offer honest medical information to parents, families, investigators
- Mandated reporter
- Provide medical home
- Provide anticipatory guidance, assess for psychosocial risk factors
- NOT to blame caregivers or investigate criminal activity

DEPENDENCY COURT

- Where can the child live safely?
- County counsel
- Requires preponderance of the evidence
  - Court must have >50% proof that abuse or neglect has occurred
- In most cases, goal is reunification
  - Offer support/services to family

CRIMINAL COURT

- State attempts to prosecute perpetrator where there has been serious bodily harm or death
- District Attorney
  - Complex Child Abuse Section
- Standard of proof
  - Beyond a reasonable doubt
  - Court must have >99% proof that an abusive event has occurred

VIOLENCE INTERVENTION PROGRAM

- Open 24/7, located in the Outpatient Building at LAC+USC Medical Center
- Provides forensic and medical evaluations for children involved with DCFS
  - Alexis Project for LGBTQ+ youth
  - Fetal Alcohol Spectrum Disorder clinic
- Also work directly with law enforcement
  - SART center for sexual assault patients
- Have mental health and case management on-site in clinic
- Separate mental health campus for on-going mental health services
- Social assistance
  - Need for short-term hotel for DV victims, need for furniture/clothing, taxi vouchers, market gift cards, weekly food drive, etc.

RESOURCES AT KAISER

- Child abuse champions – Regional Lead
  - **Mercie DiGangi, Downey**
- Child abuse champions – Local

- **Rashmi Dayal, Antelope Valley**
- **Lisa Richie, Baldwin Park**
- **Tintin Manuela-Abad, Kern County**
- **Evan Tarragano, Los Angeles**
- **Altaf Kazi, Orange County**
- **Amanda Green, Panorama City**

- **Jenna Nolan, Riverside**
- **Allison Nguyen, San Bernadino County**
- **Sara Valladolid, San Diego**
- **Sara Stewart, South Bay**
- **Amanda Weiler, Woodland Hills**
- **Elizabeth Azinge, West LA**

- TBA – Upcoming partnership with VIP and Kaiser

KAISER ORDER SET

MEDICATIONS - Adolescents - Right Click to Edit Details

- Medications - Plan B - Back Office — Click for more
- Medications - Chlamydia — Click for more
- Medications - Gonorrhea - CHECK ADMINISTRATION PROCEDURE AND DIAGNOSIS CODE — Click for more
- Medications - Herpes Simplex Virus — Click for more
- Medications - HPV (Human Papilloma Virus) — Click for more
- Medications - Syphilis — Click for more
- Medications - Trichomonas — Click for more

ORDERS - Right Click for Order Detail, Future, Standing, etc.

- Laboratory - Hematologic/Bruising Screen - SELECT Dx AND Lab Order — Click for more
- Laboratory - Abdominal Trauma Screen - SELECT Dx AND Lab Order — Click for more
- Laboratory - Rickets Screen - SELECT Dx AND Lab Order — Click for more
- Laboratory - Sexual Abuse Related Work-Up - SELECT Dx AND Lab Order - REFER ALL ACUTE PATIENTS TO FORENSIC CENTER — Click for more
- Imaging — Click for more
- Procedures — Click for more
- Referrals - List below is for clinical guidance. SOCIAL SERVICES should ALWAYS be requested. — Click for more

FOLLOW-UP

- Follow up - Right Click to Edit Disposition and Follow up Data — Click for more
  - FOLLOW-UP APPT WITH PRIMARY CARE PROVIDER

PATIENT INSTRUCTIONS

- Patient Instructions - Click [+] for Additional Patient Instruction Choices — Click for more
  - AFTERCARE INSTRUCTIONS SCAN PEDS SCAL [172861]

CONCLUSIONS

- Have Abusive Head Trauma on your differential when there is head trauma with no history of trauma or history of trauma is inconsistent with injuries in a young infant
- The diagnosis of Abusive Head Trauma requires a comprehensive, multidisciplinary assessment and has solid backing in medical literature and research
- Bruising in non-ambulatory children is a common sentinel finding and needs to be reported
- Ask for help –
  - Use your Child Abuse Champions to assist in your evaluation
  - And, hopefully soon, VIP at LAC+USC !!

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